Investigation of the Use of Neural Networks for Anti-Skid Brake System Design

Sanjay K. Mazumdar[‡] and Cheng Chew Lim[§]
[‡]Weapons Systems Division, Defence Science Technology Organisation (Australia)
PO Box 1500, Salisbury South Australia 5108
[‡] [§]Department of Electrical & Electronic Engineering
The University of Adelaide, South Australia 5005

Abstract

A neural network based model reference adaptive control approach to anti-skid brake system (ABS) design is investigated in this paper. The principal benefit of using neural networks in an ABS is their ability to adapt to changes in the environmental conditions without significant degradation in performance. In the proposed approach, the controller neural network is designed to produce a braking torque which regulates the wheel slip for the vehicle–brake system to a prespecified level. Simulation studies are performed to demonstrate the effectiveness of the proposed neural network based anti-skid brake system (NN-ABS).

1 Introduction

The objective of an anti-skid brake system (ABS) is to maximise tyre traction by preventing the wheels from locking during braking whilst maintaining adequate vehicle stability and steerability. Basically an anti-skid brake system comprises sensors to detect imminent wheel-locking together with a system to modulate the applied brake pressure so that the braked wheel maintains a tangential speed slightly less than the linear speed of the vehicle. This enables the tyres to retain most of their lateral force capability which in turn helps to keep the vehicle steerable and the overall system stable. Furthermore, the stopping distance of a vehicle equipped with an ABS is, in most cases, reduced in comparison with a stop achieved by wheel-locking.

The anti-skid brake system is a challenging control problem because the vehicle–brake dynamics are highly nonlinear with uncertain time-varying parameters. These parameter variations are due to factors such as changes in the brake pad coefficient of friction, changes in the gradient of a road, and variations in the friction characteristics of the tyre/road contact. The latter variations are perhaps the most significant and are produced by differences in the methods of road surface construction, contaminants on the road, changes in tyre parameters such as tread wear and variations in the speed of the vehicle [2, 5, 6]. Currently most commercial anti-skid brake systems are based on a look-up table approach. These look-up tables are calibrated through iterative laboratory experiments and engineering field tests. Therefore, these systems are not adaptive and issues such as robustness are not addressed. More recently some theoretical studies of anti-skid brake systems have begun to appear in the literature [1, 5, 6, 7]. In particular, control approaches such as sliding control, describing function methods, optimal control and fuzzy logic control have been used. In many of these approaches, the effects of the above parameter variations have been considered. It will be shown that the proposed neural control scheme possesses the properties described above. In particular, the adaptive nature of the approach will help to ensure that the performance of the proposed neural network based anti-skid brake system (NN-ABS) is still satisfactory even when adverse road conditions are encountered.

2 Derivation of the System Dynamics

The simplified model of the vehicle–brake dynamics presented in [5] is employed in this study. The mathematical model consists of single wheel rotational dynamics, linearised vehicle dynamics and the interactions between them. The actuator dynamics, wind resistance effects and all the dynamics associ-

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ated with the suspension system and steering mechanism are not considered.

The differential equation describing the vehicle dynamics is given by

$$\dot{\omega}_v(t) = -\frac{1}{M_v R_w}[n_w F_t(t) + B_v R_w \omega_v(t) + F_\theta(t)] \quad (1)$$

where $F_\theta(t) = M_v g \sin(\theta(t))$ is the force applied to the car as a result of a vertical gradient in the road and $F_t(t) = \mu(\lambda) N_v(\theta(t))$ is the tyre friction force which is a function of the tyre normal force $N_v(\theta(t)) = \frac{M_v}{n_w} g \cos(\theta(t))$. The equation for the normal force assumes that the weight of the car is distributed evenly amongst all the wheels. The dynamics of the wheel are described by the following differential equation

$$\dot{\omega}_w(t) = \frac{1}{J_w}[-T_b(t) - B_w \omega_w(t) + T_t(t)] \quad (2)$$

where $T_t(t) = R_w F_t(t)$ is the torque generated due to the slip between the wheel and the road surface. The most significant variable in the vehicle–brake system is the wheel slip, λ. For a braking operation, the wheel slip is governed by

$$\lambda(t) = \frac{\frac{V_v(t)}{R_w} - \omega_w(t)}{\frac{V_v(t)}{R_w}} = \frac{\omega_v(t) - \omega_w(t)}{\omega_v(t)} \quad (3)$$

Wheel slip is usually chosen as the controlled variable in anti-skid braking systems because of its influence on the tyre frictional force F_t, and its corresponding torque, T_t. This influence arises due to the relationship between the wheel slip and the adhesion coefficient (or braking friction coefficient), μ. The nonlinear relationship between the adhesion coefficient and the wheel slip is represented by a $\mu - \lambda$ characteristic. These characteristics are strongly influenced by both the material and method of construction of the road as well as tyre parameters such as the tread wear. Factors such as the vehicle speed and vertical load also have an influence on this characteristic. The adhesion coefficient - wheel slip characteristics for four common road surfaces are given in Figure 1. These characteristics are based on experimentally measured $\mu - \lambda$ data provided in [2].

Based on the $\mu - \lambda$ characteristics, the objective of an ABS is to regulate the wheel slip so as to maximise the adhesion coefficient for a given road surface. However to ensure that the system does not move into the unstable (-ve slope) region of operation, a slightly more conservative value of desired slip is chosen by many ABS engineers. Typically the vehicle–brake system is made to operate within its positive slope (stable) $\mu - \lambda$ region and it is regulated to the slip ($\lambda_{\mu_{max}}$) which produces maximum braking friction only when a maximum braking force is required. For the four surfaces considered in this study, a conservative value of slip of $\lambda = 0.15$ is used.

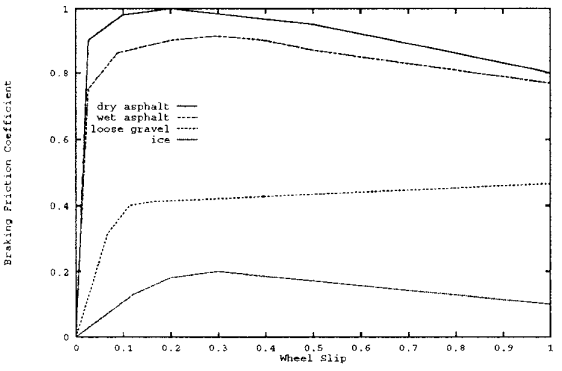


Figure 1: Adhesion coefficient (μ) vs wheel slip (λ) for various road surfaces

As the neural controller is based on discrete-time systems, the above differential equations are discretised using an Euler discretisation process. This results in the following difference equations:

$$\begin{aligned}\lambda((k+1)h) = (2 &- \frac{\omega_v((k+1)h)}{\omega_v(kh)})\lambda(kh) \\ &- \frac{\omega_w((k+1)h) - \omega_w(kh)}{\omega_v(kh)} \\ &+ \frac{\omega_v((k+1)h)}{\omega_v(kh)} - 1 \end{aligned} \quad (4)$$

with

$$\omega_v((k+1)h) = (1 - a_2 h)\omega_v(kh) - a_1 h \mu(\lambda(kh)) - a_0 h \quad (5)$$

$$\begin{aligned}\omega_w((k+1)h) &= (1 - b_2 h)\omega_w(kh) - b_1 h T_b(kh) \\ &+ b_0 h \mu(\lambda(kh)) \end{aligned} \quad (6)$$

where $a_2 = \frac{B_v}{M_v}$, $a_1 = \frac{n_w N_v(\theta(t))}{M_v R_w} = \frac{g n_w \cos(\theta(t))}{n_w R_w}$, $a_0 = \frac{F_\theta(t)}{M_v R_w} = \frac{g \sin(\theta(t))}{R_w}$, $b_2 = \frac{B_w}{J_w}$, $b_1 = \frac{1}{J_w}$, $b_0 = \frac{R_w N_v(\theta(t))}{J_w} = \frac{M_v g R_w \cos(\theta(t))}{n_w J_w}$, h is the sampling period, the control is the braking torque, T_b, the measurable states of the system are $\omega_v((k+1)h)$ and $\omega_w((k+1)h)$, and the overall system response is the slip, $\lambda((k+1)h)$. It is this quantity which will be controlled by the neural network based anti-skid brake system.

3 Neural Network Based Anti-skid Brake System

3.1 System Description

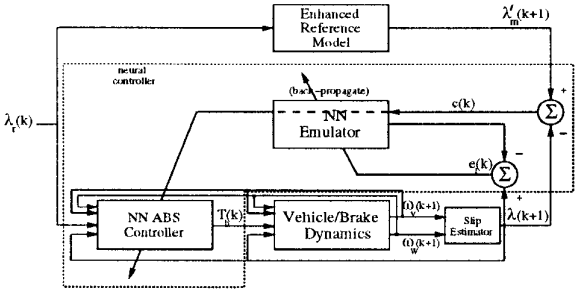


Figure 2: Block diagram of the neural network anti-skid brake system

A block diagram of the neural network based anti-skid brake system is shown in Figure 2. The inputs to the neural network adaptive controller are the wheel slip (λ), vehicle speed (ω_v), wheel speed (ω_w) and reference input (λ_r). The output from the NN-ABS controller is the braking torque (T_b). It is assumed that if the system is implemented on an actual vehicle, wheel sensors would be available to provide measurement of the angular wheel speed (ω_w) and either an estimation scheme or a sensor such as the Doppler ground radar would provide a value of linear vehicle speed (V_v) from which the equivalent angular speed ($\omega_v = \frac{V_v}{R_w}$) could be obtained. Therefore, in the NN-ABS scheme, it is assumed that the measurable system parameters are the equivalent angular vehicle speed and the angular wheel speed. The value of slip is estimated via equation (4). The value of braking friction coefficient (μ) used in the vehicle–brake model, equations (5) and (6), is calculated from the wheel slip using the $\mu - \lambda$ characteristic provided in Figure 1. The wheel slip is assumed to be the overall output response. Therefore the neural network emulator models the entire system dynamics, i.e., the vehicle–brake dynamics and the wheel slip dynamics. The neural network based anti-skid brake system is described as follows:

The overall system output (wheel slip) is governed by the following nonlinear difference equation

$$\lambda((k+1)h) = f(\omega_v(kh), \omega_w(kh), T_b(kh), \lambda(kh)) \quad (7)$$

where $f(.)$ is an arbitrary smooth nonlinear function representing the relationships defined by equations (4), (5) and (6), and T_b is the control variable.

Consider a stable reference model

$$\lambda_m((k+1)h) = f_m(\lambda_m(kh), \lambda_r(kh)) \quad (8)$$

where $f_m(.)$ is a stable linear function. The aim of the control is to find a suitable braking torque (T_b) such that the wheel slip (λ) tracks the model output (λ_m). The control strategy is to choose a $T_b^*(kh)$ such that the corresponding $\lambda^*((k+1)h)$ is nearest, in some norm sense, to a desired output $\lambda'_m((k+1)h)$. This output variable is to be specified *a priori* as a function of the wheel slip and the model output. An appropriate way to generate $\lambda'_m((k+1)h)$ is

$$\lambda'_m((k+1)h) = \lambda_m((k+1)h) - \beta(\lambda_m(kh) - \lambda(kh)) \quad (9)$$

where $\beta < 1$ is a constant. The combination of the reference model, equation (8), and the enhanced reference output, equation (9), is called the enhanced reference model in Figure 2.

By minimising the cost function given by

$$E = \frac{1}{2}\sum_{k=0}^{N-1}(\lambda'_m((k+1)h) - \lambda((k+1)h))^2 \quad (10)$$

where N is the number of samples, a suitable braking torque $T_b(kh)$ is generated.

The motivation for using $\lambda'_m((k+1)h)$ in the cost function is that a sufficient condition to ensure the convergence of the tracking error can be established. To explain, define the tracking error as

$$e_T((k+1)h) = \lambda_m((k+1)h) - \lambda((k+1)h). \quad (11)$$

In addition, a residual is defined as

$$c((k+1)h) = \lambda'_m((k+1)h) - \lambda((k+1)h) \quad (12)$$

which represents the difference between the enhanced output and the slip. It follows that

$$e_T((k+1)h) = \beta e_T(kh) + c((k+1)h) \quad (13)$$

During the computation of the control variable $T_b(kh)$, if $c((k+1)h)$ is made to satisfy the inequality

$$\|c((k+1)h)\| \leq \frac{1-\beta}{\beta}\|e_T(kh)\| \quad (14)$$

where $\|.\|$ is the l_2-norm, then the tracking error e_T converges to zero as $k \to \infty$. A mathematical justification for this condition is provided in Appendix A.

Minimising the cost function given in (10) gives rise to the following controller weight update equation

$$W_c((k+1)h) = W_c(kh) + \eta e_c(kh)\frac{\partial T_b(kh)}{\partial W_c(kh)} \quad (15)$$

where η is the learning rate and the controller error is given by

$$\epsilon_c((k+1)h) = (1-\beta)c((k+1)h)\frac{\partial \hat{\lambda}((k+1)h)}{\partial T_b(kh)} \quad (16)$$

The term $\frac{\partial \hat{\lambda}((k+1)h)}{\partial T_b(kh)}$ is an approximate of the plant Jacobian obtained by backpropagating through the neural network emulator [4] which is defined by the following equation

$$\hat{\lambda}((k+1)h) = \hat{f}(\omega_v(kh), \omega_w(kh), T_b(kh), \lambda(kh); W_I) \quad (17)$$

where $f(.)$ is a multilayered neural network parameterised by the weights W_I. Therefore, the identification of the vehicle-brake dynamics is only necessary to obtain an approximate of the plant Jacobian which is then used to update the controller weights. The weight update equation (15) is then used in combination with the backpropagation algorithm to synthesise the desired control. An iterative search scheme similar to the one proposed in [3] is used in conjunction with the above weight update equation to generate a control which ensures that the residual satisfies the inequality (14).

3.2 Simulation Results

In the simulation study, neural networks consisting of 4 inputs, 2 hidden layers with 25 and 20 nodes, respectively, and 1 output are used. A learning rate of $\eta = 0.02$ is used for both neural networks. The reference model used is given by the following difference equation

$$\lambda_m((k+1)h) = (1-20h)\lambda_m(kh) + 20h\lambda_r(kh) \quad (18)$$

where h is the sampling period. As can be seen, the reference model is a linear system with unity gain, so that the desired level of wheel slip is represented by the reference input $\lambda_r = 0.15$. The reference model is chosen to have a quick response time, thus ensuring that the reference slip value rapidly reaches the desired slip. The enhanced reference output is chosen to be

$$\lambda'_m((k+1)h) = \lambda_m((k+1)h) - 0.1(\lambda_m(kh) - \lambda(kh)) \quad (19)$$

The sampling period is chosen to be $h = 1$ms. Fast sampling is used because generally in an ABS the changes in the system take place very rapidly, i.e., in the order of one second. The values of the various vehicle-brake system constants are $B_w = 4$Ns, $B_v = 6$Ns, $J_w = 1.13$Nm, $M_v = 4 \times 342$kg, $R_w = 0.33$m and $n_w = 4$.

The first simulation scenario considered is the performance of the neural network based anti-skid brake system for dry asphalt, wet asphalt, loose gravel and icy road surfaces. The vehicle is travelling on a level surface ($\theta = 0$) and transitions between the road surfaces are not considered. The ABS is initiated when the vehicle speed is 60 km/h and the wheel slip is 0.5. This is a typical value of wheel slip at which the ABS is initiated, because the wheels are tending towards locking-up and, on most surfaces, this value corresponds to the unstable (-ve slope) region on the $\mu - \lambda$ characteristic.

An indication of the effectiveness of the approach can be obtained by comparing the braking distance for the NN-ABS with the braking distance obtained from a locked-wheel stop (Table 1). The braking distance is defined as the distance travelled by the vehicle once the brakes have been applied until the vehicle stops. As can be seen from these results

road surface	ABS	lock-up	Δ_{lock} %
dry asphalt	14.24	17.60	-19.1
wet asphalt	15.94	18.33	-13.0
loose gravel	34.20	30.06	+11.2
ice	91.83	134.67	-31.8

Table 1: Braking distances (metres) for the neural network anti-skid brake system versus a locked-wheel stop for an initial vehicle speed of 60 km/h

the braking distances are reduced appreciably when the anti-skid brake system is employed in all of the cases except for the loose gravel road surface. This is reflected in the values for the percentage decrease in braking distance for the NN-ABS in comparison with the results for a locked-wheel stop (Δ_{lock}). However, for a loose gravel surface, maximum adhesion occurs at $\lambda = 1$, and thus a minimum braking distance is achieved when a locked-wheel stop is employed. Therefore the results obtained for this case are as expected.

The results for the dry asphalt – loose gravel road transition (Figure 3) demonstrate that the NN-ABS approach is capable of dealing with relatively drastic changes in road conditions with minimal effect on

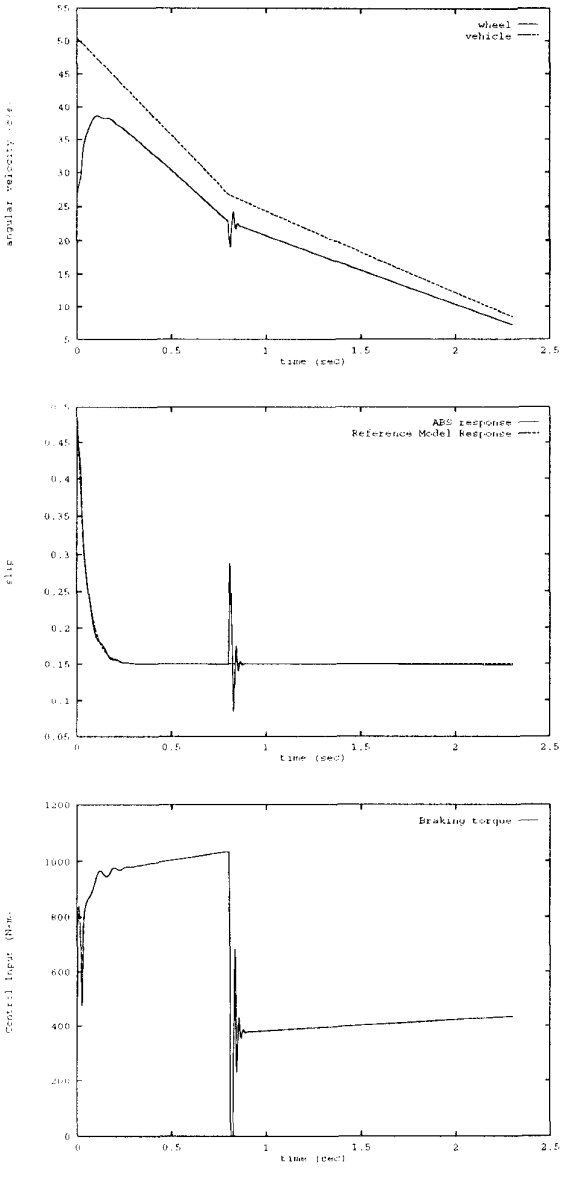


Figure 3: Angular velocity, wheel slip and braking torque responses for a dry asphalt - loose gravel transition

its ability to produce the desired slip response. In fact apart from a short oscillation (< 0.1s) which occurs when the transition takes place, the slip closely tracks the desired response. The braking torque results for the road surface transition simulation is comprised of two distinct regions. The first region consists of a braking torque of a relatively large magnitude, while after the transition occurs, the braking torque is much smaller. This can be explained by the fact that in this simulation, the vehicle moves from a surface with a relatively large coefficient of braking friction (dry asphalt) to a surface which does not offer much frictional resistance (loose gravel). Therefore, a large braking torque can initially be applied without lock-up occurring, but once the transition takes place, the braking torque must fall in order to maintain the desired level of slip on the reduced friction surfaces.

4 Conclusions

In this paper, a neural network adaptive control scheme is applied to the practical problem of an automotive anti-skid brake system. The principle objective of the proposed neural network based anti-skid brake system is to maximise the tyre traction during a braking operation by regulating the vehicle–brake dynamics to a suitable level of slip for any given road surface. In this approach, the controller neural network is designed to produce the braking torque required to ensure that the vehicle–brake system produced a wheel slip which tracked the desired slip response characterised by a reference model. The simulation study undertaken shows that the neural network based anti-skid brake system is able to produce effective braking responses for the vehicle even when it is subject to severe environmental conditions.

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Appendix A : Derivation of the sufficient condition for the convergence of the tracking error

For the sake of generality, a multi-input multi-output (MIMO) system will be considered in this appendix. Furthermore, for the sake of simplicity, a sampling period of $h = 1$ will be used. Consider the tracking error

$$e_T(k+1) = Ae_T(k) + c(k+1); \qquad e_T(0) = e_0 \quad \text{(A.1)}$$

where A is a Hurwitz (stable) matrix. The aim is to show that if $c(k+1)$ satisfies the inequality

$$\|c(k+1)\| \leq \frac{1-\lambda_{max}}{K} \|e_T(k)\| \quad \text{for } k \in \mathbb{N} \quad \text{(A.2)}$$

where $\|.\|$ is the l_2-norm, $K = \|S\| \, \|S^{-1}\|$, λ_{max} is the largest eigenvalue of A and $A = S\Lambda S^{-1}$, then $\|e_T(k)\| \to 0$ as $k \to \infty$. The solution of equation (A.1) takes the form

$$e_T(k) = A^k e_0 + \sum_{i=1}^{k} A^{k-i} c(i) \quad \text{(A.3)}$$

Taking l_2-norms of both sides of equation (A.3) yields

$$\|e_T(k)\| \leq \|A^k\| \, \|e_0\| + \sum_{i=1}^{k} \|A^{k-i}\| \, \|c(i)\| \quad \text{(A.4)}$$

Assume that A is chosen such that it has real and distinct eigenvalues. Therefore

$$A^k = S\Lambda^k S^{-1} \quad \text{(A.5)}$$

where Λ is a diagonal matrix with the eigenvalues of A on the main diagonal and S is the corresponding matrix of eigenvectors of A. It follows that

$$\begin{aligned} \|A^k\| &\leq \|S\| \, \|\Lambda^k\| \, \|S^{-1}\| \\ &\leq K\lambda_{max}^k \end{aligned} \quad \text{(A.6)}$$

where $K = \|S\| \, \|S^{-1}\|$ and λ_{max} is the largest eigenvalue of A. From (A.4) and (A.6), one gets

$$\|e_T(k)\| \leq K\lambda_{max}^k \, \|e_0\| + \sum_{i=1}^{k} K\lambda_{max}^{k-i} \, \|c(i)\| \quad \text{(A.7)}$$

Consider that inequality (A.2) holds. Therefore (A.7) becomes

$$\|e_T(k)\| \leq K\lambda_{max}^k \, \|e_0\| + \sum_{i=1}^{k} m\lambda_{max}^{k-i+1} \, \|e_T(i-1)\| \quad \text{(A.8)}$$

where $m < \frac{1-\lambda_{max}}{\lambda_{max}}$. Multiplying (A.8) by λ_{max}^{-k} and defining $\gamma(k) = \lambda_{max}^{-k}\|e_T(k)\|$, yields

$$\gamma(k) \leq K \, \|e_0\| + \sum_{i=1}^{k} m\gamma(i-1) \quad \text{(A.9)}$$

(A.9) can further be written as

$$\gamma(k) \leq [K \, \|e_0\| + m\gamma(0)](m+1)^{k-1} \quad \text{(A.10)}$$

With $\gamma(0) = \|e_0\|$

$$\gamma(k) \leq (K+m) \, \|e_0\| \, (m+1)^{k-1} \quad \text{(A.11)}$$

Multiplying (A.11) by λ_{max}^k, one gets

$$\|e_T(k)\| \leq (K+m) \, \|e_0\| \, (m+1)^{k-1}\lambda_{max}^k \quad \text{(A.12)}$$

or

$$\|e_T(k)\| \leq \frac{K+m}{m+1} \, \|e_0\| \, [(m+1)\lambda_{max}]^k \quad \text{(A.13)}$$

As $m < \frac{1}{\lambda_{max}} - 1$, then $(m+1)\lambda_{max} < 1$. Thus as $k \to \infty$, $\|e_T(k)\| \to 0$.

For a single-input single-output system, such as the anti-skid brake system considered in this paper, $A = (\beta) \in \mathbb{R}^{1\times 1}$, and thus $\lambda_{max} = \beta$ and $K = \beta$.